

Never Type A Total

A course exemplar I had published read $10 \times \$8.44 = \80.44 . It is wrong by nearly four dollars. It sat there through my own proofreading and through a class of students, **because a printed number looks like a fact.**

The fix is not "check more carefully."

The fix is that **the total is never typed at all**. It is worked out from the parts every time it is shown, so the wrong number is not something you have to catch. It is something that cannot be written down.

Where you already type totals

Answer keys. Points possible at the top of a test. A materials list with a cost at the bottom. A field trip letter home. A rubric that is supposed to add to 100. A price list on the wall. **Every one of those is a number somebody typed once and now has to maintain by hand**, and every one of them lies silently the moment a row changes.

Two rules, and the second one surprises people

- 1 · **Compute from the parts.** If a number can be worked out from other numbers on the page, it is never typed.
- 2 · **Keep money in whole cents.** Computers get decimal money wrong in a way that is genuinely surprising, and page 3 shows it happening rather than claiming it.

Where this comes from

I teach high school woodshop and I built the software my own classes use. **Both rules are in it because I got them wrong first**, and the eighty-dollar total above is the mistake that caused the rewrite.

RULE 1

Compute it from the parts, every time

The rule in one line: **if a number can be worked out from other numbers on the page, you do not type it.** Not into a document, not into a spreadsheet cell, not onto a poster.

Why proofreading does not catch it, and this is the whole problem

A wrong total looks approximately right. \$80.44 is in the neighborhood of \$84.40. The eye checks the shape of a number, not its value, and a transposed digit keeps the shape.

And nobody re-checks a number they typed themselves. You worked it out once, carefully, and from then on it is furniture. The second read is reading your own earlier confidence.

The part that actually bites, months later

The typo is not the real danger. **The real danger is the change.** A price goes up, a question gets dropped, a rubric row is reworded and reweighted. The parts move and **the typed total stays exactly where it was**, now confidently wrong, and nothing on the page indicates that anything happened.

What this looks like with no software at all

In a spreadsheet: the total cell is a formula. Never a number. If you have ever typed a figure into a cell under a column of numbers, that is the habit this page is about.

In a document: if the total cannot be a formula, **do not print it.** A materials list with no total is correct forever. A materials list with a stale total is worse than useless, because people trust it.

On a rubric: state the weights, and let the points be the sum of the rows. The "out of 100" at the top is the single most commonly wrong number in a teacher's documents, and it is wrong because somebody reweighted a row in August.

Build it so the wrong thing is impossible, rather than so that you are careful.

This is the same move as not having a box to type a student's name into, and the same move as giving every student a differently ordered test instead of policing copying. **You do not remember a rule. You remove the way to break it.**

RULE 2

Keep money in whole cents

Here is the part that catches people who did everything else right. **Computers do not store decimal money exactly**, so even a computed total can come out wrong-looking. These are real outputs, not an illustration.

WHAT A COMPUTER ACTUALLY RETURNS

$0.1 + 0.2$	→	0.300000000000000004
8.44×10	→	84.39999999999999
1.10×3	→	3.3000000000000003
$2.35 + 1.10 + 0.95 + 4.04$	→	8.440000000000001

Why this matters even though it is a tiny amount

Nobody loses money over four zeros and a one. **What you lose is trust**, in front of a class, when the board says \$8.440000000000001 and thirty students decide the software is broken. And the moment you round it for display, you have two different numbers in the system and no rule about which one is real.

The fix, and it is one sentence

Store money as a whole number of cents and divide only when you show it. \$8.44 is 844. Ten of them is 8,440. Shown, that is \$84.40, exactly, every time.

The same four parts above, added as cents and divided at the end, give **8.44** with nothing after it. The arithmetic did not change. **What changed is that the rounding happens once, at the end, where you can see it.**

And the one that caught me: fractions typed as decimals

A shop runs on fractions, and so does a recipe, a sewing pattern and a scale drawing. **A board is three-quarters of an inch, not 0.75**, and the moment somebody retypes it as 0.75 you have lost the thing the student was supposed to read. Keep the fraction the student sees, **parse it for the arithmetic**, and never make a child read a decimal of an inch because the software found it easier.

If you are not writing software, the same rule reads:

round once, at the end, where somebody can see it happen. Not in the middle of a column, not differently in two places, and never twice.

Six places to look, and what to do about each

Go through the materials you already use. **You are looking for numbers you typed that something else could work out.** This is quicker than it sounds and it rarely comes up empty.

#	Where to look	What to do
1	Points possible at the top of a test or a rubric	Make it the sum of the rows, or delete it.
2	Answer keys where an answer is arithmetic on the question	Work every one again from the question, not from the key.
3	Materials and supply lists with a cost at the bottom	A formula, or no total at all.
4	Letters home quoting a price or a date count	Check against the source the day you send it.
5	Anything you copied from last year	Every number in it is a year old. Treat it as unverified.
6	A total that appears twice in one document	Delete one of them. Two copies of a number is two chances to be wrong.

The one nobody expects to find

Row 6. A figure in the body and the same figure in a summary box is the most reliable source of contradiction in any document, because only one of them ever gets updated. **If you find a document where the two disagree, you have found out which one people actually read.**

And if you ask AI to help with any of this

It will happily add a column for you and it is often right. **It is also perfectly willing to produce a total that is confidently wrong**, in the same calm voice, and it will not tell you which kind of answer you just got. **Ask it for the formula rather than the answer.** A formula you can check in two seconds; a number you have to redo by hand.

Find one typed total in something you already use

Pick a document you hand out or hang up. Not your best one — an ordinary one you have used for more than a year, because age is what makes a typed total go wrong.

The document

The number I did not work out again

Write the number, and where on the page it sits.

Work it out from the parts now. What did you get?

☐ They match ☐ They do not ☐ I cannot tell what the parts are, which is its own finding

What I am going to do so it cannot happen again

A formula, a deletion, or a note saying where the real number lives. **"Be more careful" is not an answer** and will not survive August.

And how long had it been wrong? If the document was a copy from last year, the honest answer is a year.

If you would rather the rule lived in a tool than in your memory

Our **Vibe Coding for Teachers** build-alongs hand you a working classroom tool you own outright, with your own content going in from a setup screen and no code to edit. **A total a tool computes is a total you never have to check again.** Search our store.

If this was useful, would you leave a review?

TPT gives you credit toward future purchases, and reviews decide whether another teacher ever sees this. Every figure on page 3 is real output, not an illustration.